

Practical - VII

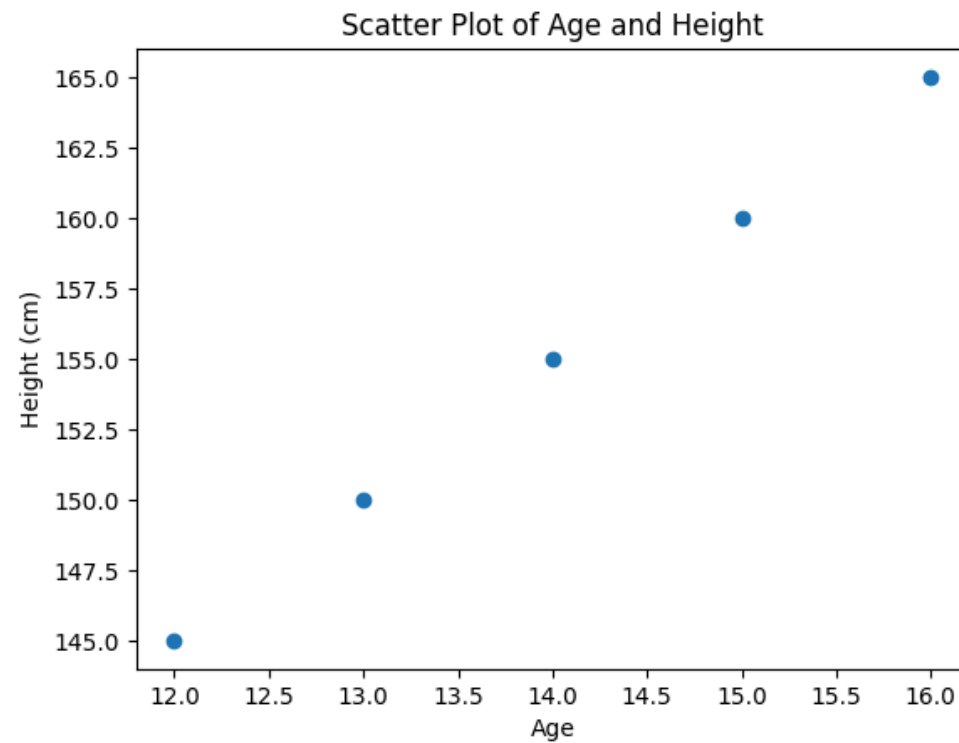
Roll No. :S040	Name:Aditya Sopte
Class :SYIT	Batch :
Date of Assignment :02-09-2007	Date/Time of Submission :02-09-2007 ☐

a. Write a Python program to create a scatter plot of Age and Height of 5 students.

```
import matplotlib.pyplot as plt
```

```
age = [12, 13, 14, 15, 16]  
height = [145, 150, 155, 160, 165]
```

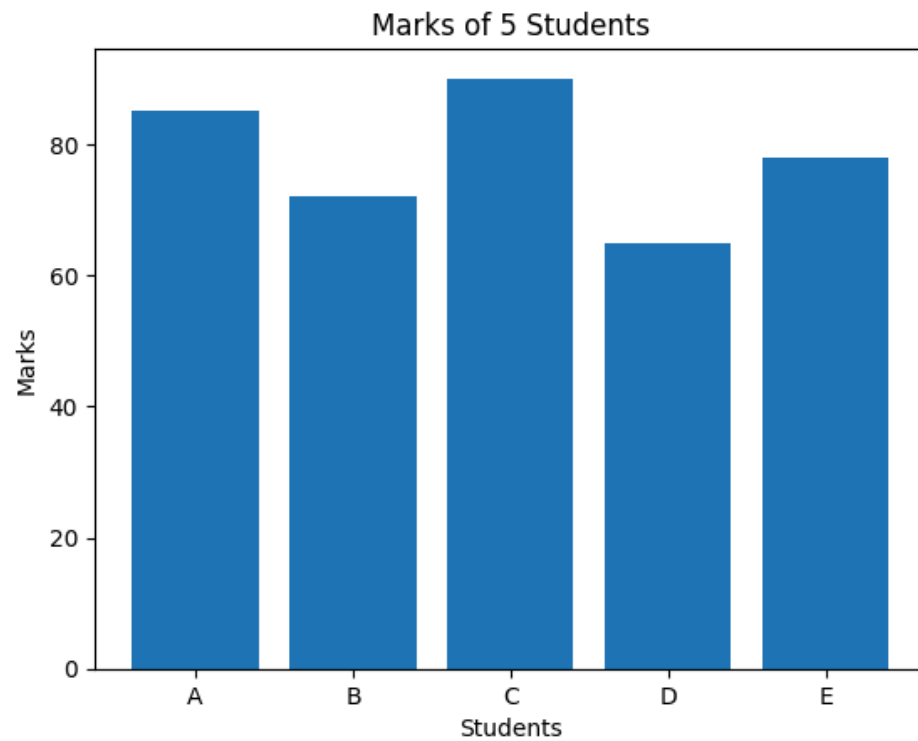
```
plt.scatter(age, height)  
  
plt.xlabel("Age")  
plt.ylabel("Height (cm)")  
plt.title("Scatter Plot of Age and Height")  
  
plt.show()
```



b. Write a Python program to create a bar chart showing marks of 5 students.

```
students = ["A", "B", "C", "D", "E"]  
marks = [85, 72, 90, 65, 78]
```

```
plt.bar(students, marks)  
  
plt.xlabel("Students")  
plt.ylabel("Marks")  
plt.title("Marks of 5 Students")  
  
plt.show()
```



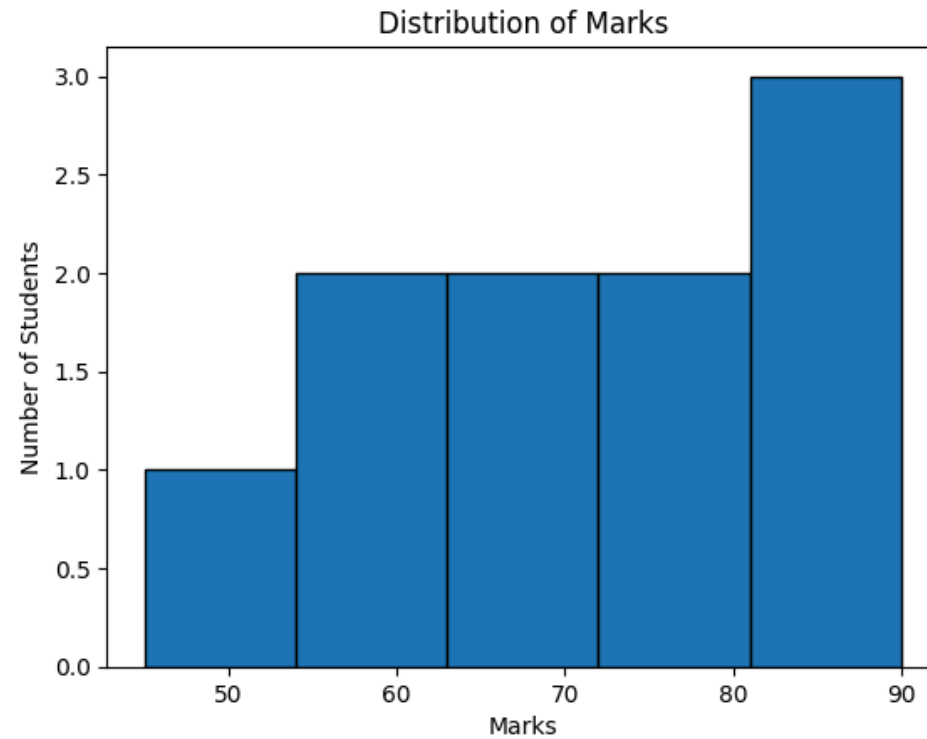
c. Write a Python program to create a histogram showing the distribution of marks of 10 students.

```
marks = [45, 56, 67, 72, 81, 65, 90, 76, 54, 88]
```

```
plt.hist(marks, bins=5, edgecolor="black")

plt.xlabel("Marks")
plt.ylabel("Number of Students")
plt.title("Distribution of Marks")

plt.show()
```

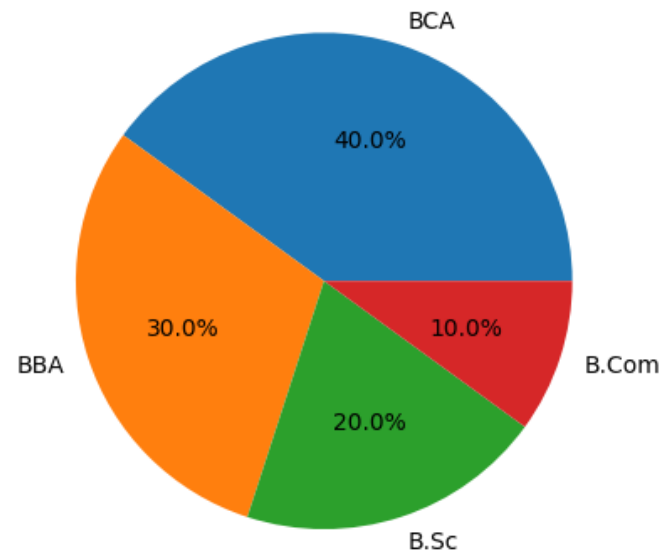


d. Write a Python program to create a pie chart showing the number of students in different courses.

```
courses = ["BCA", "BBA", "B.Sc", "B.Com"]  
students = [40, 30, 20, 10]
```

```
plt.pie(students, labels=courses, autopct="%1.1f%%")  
  
plt.title("Students in Different Courses")  
  
plt.show()
```

Students in Different Courses



e. Write a Python program to create a bar chart showing sales of 5 different products.

```
products = ["Product A", "Product B", "Product C", "Product D", "Product E"]  
sales = [120, 150, 90, 180, 130]
```

```
plt.bar(products, sales)  
  
plt.xlabel("Products")  
plt.ylabel("Sales")  
plt.title("Sales of 5 Different Products")  
  
plt.show()
```

